

Theory of Computation - 500+ One-Liners

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Table of Contents

1. Foundations and Preliminaries	5
Alphabets, Strings, Operations	5
Empty String vs Empty Language	5
Kleene Star, Plus, Sigma-star	5
Chomsky Hierarchy	5
2. Finite Automata	7
DFA Basics	7
DFA Construction Patterns	7
NFA	7
Epsilon-NFA	7
Conversions and Blow-up	7
DFA Minimization	8
Moore and Mealy Machines	8
3. Regular Expressions	9
Definition and Precedence	9
Algebraic Identities	9
Conversions	9
4. Properties of Regular Languages	10
Closure Properties	10
Pumping Lemma	10
Myhill-Nerode	10
Decision Properties	10
Regular Grammars	10
5. Context-Free Grammars	11
Definition and Derivations	11
Parse Trees and Ambiguity	11
CFG Design	11
Simplification	11
Normal Forms	11
Left Recursion and Factoring	11
6. Pushdown Automata	13
Definition	13

Acceptance Modes	13
Power and Determinism	13
7. Properties of Context-Free Languages	14
Closure	14
Pumping Lemma for CFL	14
Decision Properties	14
8. Turing Machines	15
Definition	15
Behaviour and Variants	15
UTM and Church-Turing	15
9. Recursive and Recursively Enumerable Languages	16
Definitions	16
Closure	16
10. Decidability and Undecidability	17
Decidable Problems	17
Halting Problem and Reductions	17
Rice's Theorem and PCP	17
RE-but-not-recursive Examples	17
11. Complexity (Awareness)	18
12. High-Yield Exam Facts (Rapid Fire)	19
Class vs Machine vs Grammar	19
Power Comparisons	19
State Counts	19
Canonical Languages	19
Complexity of Algorithms	19
Decidability Quick Table	19
13. Extended One-Liners (Atomic Coverage)	20
Strings and Languages (deep)	20
DFA/NFA (deep)	20
Regular Expressions (deep)	20
Closure recap (deep)	20
Pumping recap (deep)	20
CFG/PDA recap (deep)	21
TM/Computability recap (deep)	21

Decidability recap (deep)	21
Hierarchy recap (deep)	21
Numeric/quick recall (deep)	21
True/False recall (deep)	21
Mixed atomic facts (deep)	22
Examiner-favourite numeric facts	22
Final consolidation	23
14. Additional Atomic One-Liners (Set 2)	24
Foundations extended	24
DFA extended	24
NFA / conversion extended	24
Minimization extended	24
Regular expression extended	24
Closure extended	25
Pumping extended	25
CFG / PDA extended	25
CFL decision extended	25
TM extended	25
Computability extended	26
Decidability extended	26
Complexity extended	26
Master recall (Set 2)	26
15. Final Atomic One-Liners (Set 3)	28

1. Foundations and Preliminaries

Alphabets, Strings, Operations

- An **alphabet (Sigma)** is defined by exactly **2** properties: it is **finite** and **non-empty**.
- There are **2** symbols in the standard binary alphabet, namely **0** and **1**.
- A **string** has exactly **1** key length measure, written $|w|$, counting its **symbols**.
- The empty string has length **0** and is written **epsilon** (or **lambda**).
- Concatenation satisfies **1** length law: $|xy| = |x| + |y|$.
- Concatenation has exactly **1** property it lacks: it is **not commutative** (so $ab \neq ba$).
- The **identity** element for concatenation is the **1** string **epsilon**, since $w.\text{epsilon} = w$.
- A string of length **n** has exactly **n+1** prefixes, including **epsilon** and the **whole string**.
- A string of length **n** has exactly **n+1** suffixes, by the same counting.
- A string of length **n** has exactly $n(n+1)/2$ non-empty substrings, all of which are **contiguous**.
- A string of length **n** can have up to 2^n subsequences, which need **not** be contiguous.
- There are **2** distinct concepts confused as one: a **substring** (contiguous) and a **subsequence** (skips allowed).
- Reversal obeys **1** key law: $(xy)^R = y^R x^R$ (order flips).
- w^0 equals exactly **1** value: **epsilon** (any string to the power zero).
- There are **3** positional relations of a string: **prefix** (front), **suffix** (back), and **substring** (middle).

Empty String vs Empty Language

- There are **3** confusable objects: the string **epsilon**, the language $\{\text{epsilon}\}$ (size **1**), and the empty language **phi** (size **0**).
- The empty language **phi** contains exactly **0** strings.
- The language $\{\text{epsilon}\}$ contains exactly **1** string, namely **epsilon**.
- Concatenating any language with **phi** gives **1** result: **phi** (the empty language annihilates).
- Concatenating any language with $\{\text{epsilon}\}$ returns exactly **1** thing: the **original language L**.

Kleene Star, Plus, Sigma-star

- The **Kleene star L^*** always contains at least **1** guaranteed string: **epsilon**.
- The **Kleene plus L^+** contains **epsilon** only in **1** case: when **epsilon** is already in **L**.
- There is **1** relation linking them: $L^+ = L.L^*$ (and $L^* = L^+ \cup \{\text{epsilon}\}$).
- **phi**^{*} equals exactly **1** language: $\{\text{epsilon}\}$.
- **Sigma**^{*} is the set of **all** finite strings and is **countably infinite**.
- **Sigma**⁺ differs from **Sigma**^{*} by exactly **1** string: it excludes **epsilon**.
- The number of languages over an alphabet is **uncountably infinite** ($2^{\aleph_0}$), exceeding the **countable Sigma**^{*}.

Chomsky Hierarchy

- The **Chomsky hierarchy** has **4** grammar types: **Type 0**, **Type 1**, **Type 2**, and **Type 3**.
- The hierarchy maps to **4** machines: **TM**, **LBA**, **PDA**, and **Finite Automaton**.
- The hierarchy maps to **4** language classes: **RE**, **Context-Sensitive**, **Context-Free**, and **Regular**.
- **Type 0** grammars are **unrestricted** and correspond to **Turing Machines** / **RE** languages.
- **Type 1** grammars are **context-sensitive** with rule $|\alpha| \leq |\beta|$, recognised by an **LBA**.
- **Type 2** grammars are **context-free** with a **single** variable on the LHS, recognised by a **PDA**.

- **Type 3** grammars are **regular** (**left-** or **right-linear**), recognised by a **Finite Automaton**.
- The Chomsky hierarchy numbering is **inverted**: **Type 0** is the most powerful, **Type 3** the least.
- There are **4** strictly nested classes: **Regular** $\subset$ **CFL** $\subset$ **CSL** $\subset$ **RE**.
- **Noam Chomsky** proposed the hierarchy in **1956**.

2. Finite Automata

DFA Basics

- A **DFA** is defined by a **5-tuple**: $(Q, \Sigma, \delta, q_0, F)$.
- A DFA transition function has exactly **1** target: $\delta: Q \times \Sigma \rightarrow Q$.
- A DFA has exactly **1** start state, namely q_0 .
- A DFA accepts w under exactly **1** condition: $\delta^*(q_0, w) \in F$.
- A DFA accepts **epsilon** in exactly **1** case: when $q_0 \in F$.
- A DFA over k symbols with n states has exactly $n \cdot k$ transition-table entries, all of which must be **defined**.
- A **dead/trap state** has **1** defining behaviour: all its transitions **loop back** to itself and it is **non-accepting**.
- A minimal DFA has at most **1** dead state.
- The language of a DFA is **regular**, the **1** class finite automata recognise.
- There are **3** things a DFA forbids: **multiple** transitions, **epsilon** moves, and **undefined** transitions.

DFA Construction Patterns

- "Length mod $n = r$ " needs exactly n states arranged in a **cycle**.
- "Number of a's divisible by k " needs exactly k states counting **modulo k** .
- "Contains substring w " needs exactly $|w|+1$ states tracking the **longest matched prefix**.
- "Ends with string w " needs $|w|+1$ states using **prefix tracking**.
- "Binary value divisible by k " needs k states tracking the **remainder mod k** , updating as $(2 \cdot r + b) \bmod k$.
- " n -th symbol from the end is 1" needs exactly 2^n states in the minimal DFA but only $n+1$ in an NFA.

NFA

- An **NFA** is a **5-tuple** with **1** difference from a DFA: $\delta: Q \times \Sigma \rightarrow 2^Q$ (power set).
- An NFA from one state on one symbol may go to **0, 1, or many** states.
- An NFA accepts w under **1** condition: at least **1** path ends in F .
- An NFA rejects w only when **all** paths fail.
- **NFA** and **DFA** recognise exactly the **same** class: the **regular** languages.
- Non-determinism gives a finite automaton **0** extra power, only added **convenience**.
- An n -state NFA converts to a DFA with at most 2^n states.

Epsilon-NFA

- An **epsilon-NFA** adds **1** feature: transitions on **epsilon** (no input consumed).
- **epsilon-closure(q)** always contains at least **1** state: q itself.
- There are **3** equivalent finite models: **DFA**, **NFA**, and **epsilon-NFA**.
- Epsilon transitions add **0** extra language power, only **structural** convenience.

Conversions and Blow-up

- The **subset construction** converts an NFA to a DFA with worst-case 2^n states.
- An **epsilon-NFA** converts to a DFA using **epsilon-closure** plus the **subset** method.
- The DFA start state in subset construction is the **1** set **epsilon-closure($\{q_0\}$)**.
- A DFA subset is **accepting** under **1** condition: it contains at least **1** NFA **final** state.
- Converting a **DFA to NFA** is **trivial** because every DFA is already an **NFA**.

DFA Minimization

- DFA minimization yields a **unique** minimal DFA, guaranteed by the **Myhill-Nerode** theorem.
- The **table-filling** algorithm runs in $O(n^2)$ and marks **(final, non-final)** pairs first.
- **Hopcroft's** minimization algorithm runs in $O(n \log n)$, the **best known** bound.
- Minimization requires **1** preliminary step: remove **unreachable** states first.
- Two states are **distinguishable** if **some** string makes one **accept** and the other **reject**.
- The number of **Myhill-Nerode classes** equals the number of states in the **minimal DFA**.
- There are **4** steps to table-filling: remove unreachable, mark final/non-final, propagate marks, merge unmarked.

Moore and Mealy Machines

- A **Moore machine** is a **6-tuple** whose output depends on **1** thing: the **current state**.
- A **Mealy machine** is a **6-tuple** whose output depends on **2** things: **state** and **input**.
- A Moore machine emits **$n+1$** outputs for an input of length **n** .
- A Mealy machine emits **n** outputs for an input of length **n** .
- Moore and Mealy are **equivalent** in power, differing by exactly **1** output (the initial-state output).
- A Moore machine places outputs on **states**; a Mealy machine places them on **transitions**.
- Converting Mealy to Moore can need up to **$m * |O|$** states from **m** original states.

3. Regular Expressions

Definition and Precedence

- A **regular expression** has **3** base cases: **phi**, **epsilon**, and a **symbol a**.
- A regular expression has **3** operators: **union (+)**, **concatenation (.)**, and **Kleene star (*)**.
- Operator precedence has **3** levels, highest to lowest: **star**, **concatenation**, **union**.
- **Kleene's theorem** equates **3** things: **regular expressions**, **finite automata**, and **regular languages**.

Algebraic Identities

- Union has **3** basic laws: **commutative**, **associative**, and **idempotent** ($r+r=r$).
- **phi** plays **2** roles: **identity** for union ($\text{phi}+r=r$) and **annihilator** for concatenation ($\text{phi}.r=\text{phi}$).
- **epsilon** is the **1** identity for concatenation: $\text{epsilon}.r = r$.
- There is **1** star identity: $(r^*)^* = r^*$.
- phi^* simplifies to **1** expression: **epsilon**.
- There are **2** famous FALSE identities: $(r+s)^* \neq r^*+s^*$ and $(rs)^* \neq r^* s^*$.
- One key unrolling law: $r^* = \text{epsilon} + r r^*$.

Conversions

- **Thompson's construction** converts an RE of size **n** to an **epsilon-NFA** with **O(n)** states.
- **Arden's theorem** solves $R = Q + RP$ with the **1** unique solution $R = QP^*$.
- Arden's uniqueness needs exactly **1** condition: **epsilon not in P**.
- The **state-elimination** method converts a **finite automaton to an RE** by removing intermediate states.

4. Properties of Regular Languages

Closure Properties

- Regular languages are closed under all **3** Boolean operations: **union**, **intersection**, **complement**.
- Regular languages are closed under **6+** operations including **concatenation**, **star**, and **reversal**.
- Complement closure uses **1** trick: swap **final** and **non-final** states of a **complete DFA**.
- Intersection closure uses the **product construction**, giving at most **$m \cdot n$** states.
- Regular languages form a **Boolean algebra**, closed under **all** Boolean operations.
- Regular languages are closed under **homomorphism** and **inverse homomorphism**, totalling **0** exceptions among common operations.

Pumping Lemma

- The regular pumping lemma splits a string into **3** parts: **x** , **y** , **z** .
- The regular pumping lemma has **3** conditions: **$|y| \geq 1$** , **$|xy| \leq p$** , and **$xy^i z \in L$ for all $i \geq 0$** .
- The pumping lemma proves only **1** thing: **non-regularity** (it is **necessary, not sufficient**).
- **$a^n b^n$** is the classic language proven **non-regular** by the pumping lemma.
- The condition **$|xy| \leq p$** forces **y** into the **first p** symbols.

Myhill-Nerode

- The **Myhill-Nerode theorem** is the **1** test that is both **necessary AND sufficient** for regularity.
- **L** is regular iff its Myhill-Nerode relation has a **finite** number of **equivalence classes**.
- The **index** (class count) equals the size of the **minimal DFA**.
- An **infinite** index implies the language is **not regular**.

Decision Properties

- Regular languages have **4** decidable standard problems: **membership**, **emptiness**, **finiteness**, **equivalence**.
- **Membership** for a DFA is decided in **$O(|w|)$** time by **simulation**.
- **Emptiness** is decided by **1** test: whether any **final** state is **reachable**.
- **Finiteness** is decided by checking for a **cycle** on a path to a **final** state.
- **Equivalence** of regular languages is **decidable** (unlike for CFLs).

Regular Grammars

- A **right-linear** grammar uses productions of form **$A \rightarrow aB$** or **$A \rightarrow a$** .
- A **left-linear** grammar uses productions of form **$A \rightarrow Ba$** or **$A \rightarrow a$** .
- **Pure left-linear** and **pure right-linear** grammars both generate exactly the **regular** languages.
- **Mixing** left- and right-linear rules can produce a **non-regular** (context-free) language.

5. Context-Free Grammars

Definition and Derivations

- A **CFG** is a **4-tuple**: (V, T, P, S) .
- A CFG production has exactly **1** variable on the LHS, of form $A \rightarrow \alpha$.
- There are **2** canonical derivation orders: **leftmost (LMD)** and **rightmost (RMD)**.
- A **leftmost derivation** always expands the **leftmost** variable first.
- A **rightmost derivation** always expands the **rightmost** variable first.
- Each parse tree corresponds to exactly **1** leftmost and **1** rightmost derivation.

Parse Trees and Ambiguity

- A **parse tree** has **1** root (S), with **leaves** forming the **yield**.
- A grammar is **ambiguous** if some string has ≥ 2 distinct **parse trees**.
- Ambiguity is a property of the **grammar**, while **inherent ambiguity** is a property of the **language**.
- $\{a^i b^j c^k : i=j \text{ OR } j=k\}$ is a classic **inherently ambiguous** language.
- Deciding if an arbitrary CFG is ambiguous is **undecidable**.

CFG Design

- $\{a^n b^n\}$ is generated by **1** rule set: $S \rightarrow aSb \mid \epsilon$.
- Palindromes over $\{a,b\}$ use $S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$.
- Balanced parentheses use $S \rightarrow SS \mid (S) \mid \epsilon$.
- Equal numbers of a's and b's use $S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$.

Simplification

- A **useless symbol** fails **1** of **2** tests: it is **non-generating** or **unreachable**.
- Simplification removes **non-generating** symbols **before unreachable** ones.
- The recommended simplification order has **3** steps: remove **epsilon-productions**, then **unit productions**, then **useless symbols**.
- A **unit production** has **1** form: $A \rightarrow B$ (single variable RHS).
- An **epsilon-production** has **1** form: $A \rightarrow \epsilon$.
- $S \rightarrow \epsilon$ is kept only in **1** case: when **epsilon** is in the language.

Normal Forms

- **Chomsky Normal Form** allows exactly **2** production shapes: $A \rightarrow BC$ and $A \rightarrow a$.
- A CNF derivation of a length- n string uses exactly $2n-1$ steps.
- **Greibach Normal Form** requires each RHS to start with exactly **1** terminal, form $A \rightarrow a \alpha$.
- A GNF derivation of a length- n string uses exactly n steps.

Left Recursion and Factoring

- **Left recursion** $A \rightarrow A \alpha$ breaks **top-down (LL)** parsers.
- Left recursion is removed by rewriting to **right recursion**: $A \rightarrow \beta A'$, $A' \rightarrow \alpha A' \mid \epsilon$.
- **Left factoring** extracts **common prefixes** to enable **predictive** parsing.

- Removing left recursion fixes 0 ambiguity problems (they are independent issues).

6. Pushdown Automata

Definition

- A **PDA** is a **7-tuple**: **(Q, Sigma, Gamma, delta, q0, Z0, F)**.
- A PDA equals a finite automaton plus exactly **1** extra memory: a **stack** (**LIFO**, **unbounded**).
- A PDA's stack alphabet is denoted **Gamma**, with initial symbol **Z0**.
- An **instantaneous description** of a PDA has **3** parts: **state**, **remaining input**, **stack contents**.

Acceptance Modes

- A PDA has **2** acceptance modes: by **final state** and by **empty stack**.
- The **2** acceptance modes are **equivalent** for an **NPDA** (both recognise CFLs).

Power and Determinism

- **CFG** and **NPDA** are **equivalent**, both characterising the **context-free** languages.
- A **DPDA** recognises the **DCFLs**, a **proper subset** of the **CFLs**.
- **NPDA** is strictly more powerful than **DPDA** (unlike NFA vs DFA where power is **equal**).
- **{w w^R}** (even palindromes) is a CFL accepted by an **NPDA** but **not** by any **DPDA**.
- A single stack handles exactly **1** counting comparison, which is why **a^n b^n** is a CFL but **a^n b^n c^n** is not.
- **DCFLs** (by final state) are closed under **complement** but not under **union** or **intersection**.

7. Properties of Context-Free Languages

Closure

- CFLs are closed under **4** operations: **union**, **concatenation**, **star**, **reversal**.
- CFLs are NOT closed under **3** operations: **intersection**, **complement**, **difference**.
- **CFL intersect Regular** is always a **CFL** (1 stack survives).
- The non-closure of intersection is shown by $\{a^n b^n c^m\} \cap \{a^m b^n c^n\} = \{a^n b^n c^n\}$, which is **not** a CFL.

Pumping Lemma for CFL

- The CFL pumping lemma splits a string into **5** parts: **u, v, w, x, y**.
- The CFL pumping lemma pumps **2** substrings simultaneously: **v** and **x**.
- The CFL pumping lemma has **2** length conditions: $|vwx| \leq p$ and $|vx| \geq 1$.
- $a^n b^n c^n$ is proven **non-CFL** by the CFL pumping lemma but is **context-sensitive**.

Decision Properties

- CFLs have **3** decidable standard problems: **membership**, **emptiness**, **finiteness**.
- **Membership** for CFLs is decided by the **CYK algorithm** in $O(n^3)$ time using **CNF**.
- CFL **equivalence** is **undecidable**, the key contrast with regular languages.
- At least **4** CFL problems are undecidable: **equivalence**, **ambiguity**, **$L = \Sigma^*$** , and **intersection-emptiness**.

8. Turing Machines

Definition

- A **Turing Machine** is a **7-tuple**: $(Q, \Sigma, \Gamma, \delta, q_0, B, F)$.
- A standard TM transition produces **3** outputs: a **new state**, a **written symbol**, and a **move L or R**.
- A TM tape alphabet **Γ** contains the **input alphabet** plus the **blank B**.
- A TM has exactly **2** head-move directions: **Left** and **Right**.
- A TM configuration captures **3** things: **tape contents**, **head position**, **current state**.

Behaviour and Variants

- A TM can have **3** outcomes on an input: **accept**, **reject**, or **loop forever**.
- There are **5+** TM variants all equal in power: **multi-tape**, **non-deterministic**, **multi-track**, **two-way**, **enumerator**.
- A **k**-tape TM is simulated by a single-tape TM with at most **$O(t^2)$** slowdown.
- A **non-deterministic TM** adds **0** language power over a deterministic TM (only possible time speed-up).
- An **enumerator** shows a language is **recursive** iff it lists strings in **lexicographic** order.

UTM and Church-Turing

- A **Universal Turing Machine** takes **1** encoded input $\langle M, w \rangle$ and **simulates M** on w .
- The UTM is the theoretical basis of the **stored-program computer**.
- The **Church-Turing thesis** is a **thesis (not a theorem)** equating **effective computability** with **TM-computability**.
- The thesis is supported by **3** equivalent models: **TMs**, **lambda-calculus**, and **recursive functions**.

9. Recursive and Recursively Enumerable Languages

Definitions

- A **recursive** language has **1** defining machine: a TM that **halts on every input** (a **decider**).
- A **recursively enumerable** language has a TM that **accepts** members but may **loop** on non-members.
- There are **3** synonyms for RE: **semi-decidable**, **Turing-recognisable**, **recursively enumerable**.
- **Recursive** is a **proper subset** of **RE** (every recursive language is RE, not conversely).

Closure

- **Recursive** languages are closed under **complement**; **RE** languages are **not**.
- Both recursive and RE are closed under **4** operations: **union**, **intersection**, **concatenation**, **star**.
- **Post's theorem** states **L** is recursive iff **both L and L'** are **RE**.
- If **L** is RE but not recursive, then **L'** is **not even RE**.

10. Decidability and Undecidability

Decidable Problems

- **Regular** languages have at least **5** decidable problems: **membership**, **emptiness**, **finiteness**, **equivalence**, **universality**.
- **CFLs** have **3** decidable problems: **membership**, **emptiness**, **finiteness**.
- **Equivalence** is decidable for **regular** but **undecidable** for **CFL** and **TM**.

Halting Problem and Reductions

- The **Halting Problem** is **undecidable**, proven by **Alan Turing** in **1936** via **diagonalization**.
- **HALT_TM** is **RE but not recursive**.
- A **mapping reduction** $A \leq_m B$ uses **1** computable function with **x in A iff f(x) in B**.
- If $A \leq_m B$ and **B** is decidable, then **A** is **decidable** (and undecidability flows the other way).

Rice's Theorem and PCP

- **Rice's theorem** states every **non-trivial** property of a TM's **language** is **undecidable**.
- Rice's theorem applies to **semantic** (language) properties, not **syntactic** (machine) ones.
- A **trivial** property (always true or always false) is **decidable**, the **1** exception to Rice.
- The **Post Correspondence Problem (PCP)** is **undecidable** and asks for a matching **sequence of indices**.

RE-but-not-recursive Examples

- There are **3** canonical RE-but-not-recursive languages: **A_TM**, **HALT_TM**, and **L_u**.
- The **complement** of **A_TM** is **not even RE**.
- The **diagonal language** **L_d** (machines not accepting their own code) is **not RE**.
- $\{a^n b^n c^n\}$ is **recursive** (context-sensitive), unlike the undecidable TM languages.

11. Complexity (Awareness)

- **Time complexity** counts **steps** and **space complexity** counts **tape cells**, the **2** TM resources.
- The class **P** contains problems solvable in **polynomial** time by a **deterministic** TM.
- The class **NP** stands for **Non-deterministic Polynomial**, with poly-time **verifiable** certificates.
- **P subset of NP**, and whether **P = NP** is the famous **open Millennium Prize** problem.
- **NP-complete** requires **2** conditions: being **in NP** and being **NP-hard**.
- The first NP-complete problem was **SAT**, by the **Cook-Levin theorem** in **1971**.
- At least **6** classic NP-complete problems exist: **3-SAT**, **Clique**, **Vertex Cover**, **Hamiltonian Cycle**, **TSP**, **Subset-Sum**.
- **NP-hard** problems need **not** be in NP, e.g. the **Halting problem** is NP-hard but **undecidable**.

12. High-Yield Exam Facts (Rapid Fire)

Class vs Machine vs Grammar

- The **Regular** class pairs with **1** machine (**FA**) and **1** grammar (**Type 3**).
- The **CFL** class pairs with **1** machine (**PDA**) and **1** grammar (**Type 2**).
- The **CSL** class pairs with **1** machine (**LBA**) and **1** grammar (**Type 1**).
- The **RE** class pairs with **1** machine (**TM**) and **1** grammar (**Type 0**).
- There are **2** classes closed under complement: **Regular** and **Recursive** (and **CSL**); **CFL** and **RE** are **not**.

Power Comparisons

- For finite automata, **NFA = DFA** in power (determinism adds **0**).
- For pushdown automata, **NPDA > DPDA** in power (determinism loses **some** languages).
- For Turing machines, **NTM = DTM** in language power (determinism loses **0** languages).

State Counts

- "Divisible by k " needs **k** minimal-DFA states.
- " n -th from end is 1" needs **2^n** minimal-DFA states.
- "Ends with string w " needs **$|w|+1$** minimal-DFA states.
- Subset construction has worst-case blow-up of **2^n** states.

Canonical Languages

- **$a^n b^n$** is **CFL** but **not regular** (needs **1** count).
- **$a^n b^n c^n$** is **CSL** but **not CFL** (needs **2** counts).
- **ww** is **not CFL** (cannot match two halves with one stack).
- **a^p (p prime)** and **$a^{(n^2)}$** are **not regular and not CFL**.
- **$w w^R$** (palindrome) is **CFL** accepted by **NPDA** only.

Complexity of Algorithms

- **CYK** membership runs in **$O(n^3)$** time.
- DFA **membership** simulation runs in **$O(|w|)$** time.
- **Table-filling** minimization runs in **$O(n^2)$** ; **Hopcroft** runs in **$O(n \log n)$** .
- **NFA to DFA** subset construction is worst-case **$O(2^n)$** in states.

Decidability Quick Table

- **Equivalence** is decidable for **regular**, undecidable for **CFL/TM**.
- **Ambiguity** of CFGs is **undecidable**.
- **Emptiness** is decidable for **regular and CFL**, undecidable for **TM**.
- **Membership** is decidable for **regular, CFL, CSL/recursive** but only **semi-decidable** for **RE**.

13. Extended One-Liners (Atomic Coverage)

Strings and Languages (deep)

- The set of all strings of length exactly k over a c -symbol alphabet has c^k members.
- The number of strings of length $\leq n$ over a c -symbol alphabet is $(c^{n+1} - 1)/(c - 1)$.
- A **palindrome** reads the same in **2** directions: forwards and **backwards** ($w = w^R$).
- The **transpose/reversal** of **epsilon** is **epsilon**, the **1** fixed point of reversal among short strings.
- **Concatenation** of **0** copies of any string equals **epsilon**.
- A language is **finite** if it has a **finite** number of strings, else it is **infinite**.
- Every **finite** language is **regular**, requiring **0** exceptions.
- The **empty language phi** is **regular**, accepted by a DFA with **no** reachable final state.

DFA/NFA (deep)

- A DFA with n states accepting a non-empty language accepts some string of length $< n$.
- If a DFA accepts an infinite language, it accepts a string of length between n and $2n-1$ (pumping).
- A **complete** DFA defines **exactly 1** transition for **every** (state, symbol) pair.
- The **cross-product** of an m -state and an n -state DFA has up to $m \cdot n$ states for **intersection** or **union**.
- An NFA with n states has at most 2^n reachable subsets during conversion.
- The minimal DFA for **Sigma*** has exactly **1** state, which is **final**.
- The minimal DFA for **phi** has exactly **1** state, which is **non-final** (a dead state).
- A DFA recognising "**even length**" needs exactly **2** states.

Regular Expressions (deep)

- The RE $(a+b)^*$ denotes **Sigma*** over $\{a,b\}$, accepting **all** strings.
- The RE a^*b^* denotes strings with all **as** before all **bs**, totalling **1** ordering constraint.
- The RE $(a+b)^*abb$ denotes strings ending in **abb**, needing a DFA of **4** states.
- The RE $(00+11)^*$ denotes even-length strings of **matched pairs**.
- **Two** regular expressions are **equivalent** if they denote the **same** language.

Closure recap (deep)

- **Regular** languages are closed under **quotient** with any language, giving **1** more closed operation.
- **Regular** languages are closed under **substitution** by regular languages.
- **CFLs** are closed under **substitution** and **homomorphism**, but **not inverse** of all maps... actually CFLs ARE closed under **inverse homomorphism**.
- **CFLs** are closed under **union with a CFL** but not **intersection with a CFL**.
- **Recursive** languages are closed under **set difference**; **RE** languages are **not**.

Pumping recap (deep)

- The pumping length p can be taken as the number of **states** in a DFA for L .
- Pumping **down** (taking $i=0$) removes y entirely, a valid case of $xy^i z$ in L .
- The CFL pumping length relates to the grammar's **variable** count and the $2^{|V|}$ bound.
- **Ogden's lemma** is a stronger version of the **CFL pumping lemma** using **marked** positions.

CFG/PDA recap (deep)

- A **linear grammar** has at most **1** variable per RHS but it may sit in the **middle**.
- $\{a^n b^n\}$ has a **linear** grammar but no **regular** one.
- Every **regular** language is a **CFL**, requiring **0** counterexamples.
- A **DCFL** is recognised by a **DPDA** and is closed under **complement**.
- The **CYK** table for a string of length **n** has $n(n+1)/2$ cells.
- A PDA accepting by **empty stack** can be converted to one accepting by **final state** and vice versa.

TM/Computability recap (deep)

- A **decider** halts on **100%** of inputs; a **recogniser** may halt on only the **accepted** ones.
- The set of **Turing machines** is **countably infinite**, fewer than the **uncountable** set of languages.
- This counting gap proves at least **1** language is **not RE**.
- **A_TM** reduces to **HALT_TM**, both being **RE but not recursive**.
- The complement of a **recursive** language is always **recursive**.
- An **LBA** is a TM whose tape is bounded to **O(n)** (linear in input length).
- Whether an **LBA** accepts the **empty** language is **undecidable**.
- **Context-sensitive** languages are closed under **complement** (Immerman-Szelepcsényi).

Decidability recap (deep)

- "Does a DFA accept a **finite** language?" is **decidable**.
- "Are two **CFGs** equivalent?" is **undecidable**.
- "Is a **CFL** regular?" is **undecidable**.
- "Is $L(M) = \phi$ for a TM M ?" is **undecidable** by **Rice's theorem**.
- "Does TM M have exactly **5** states?" is **decidable** (syntactic, not semantic).
- "Is a given CFG **ambiguous**?" is **undecidable**.
- The **PCP** over a **1**-symbol alphabet (single letter) is **decidable**, but general PCP is **undecidable**.

Hierarchy recap (deep)

- There are exactly **4** machine classes increasing in power: **FA < PDA < LBA < TM**.
- A **regular** language is recognised at all **4** levels of the hierarchy.
- **Type 1** grammars are **non-contracting** ($|LHS| \leq |RHS|$).
- **Type 0** grammars place **0** restrictions on production form (beyond a variable on the LHS).
- **Context-free** but not **regular**: requires **exactly 1** stack to recognise.
- **Context-sensitive** but not **context-free**: requires **linear** bounded memory.

Numeric/quick recall (deep)

- A **DFA** has **5** components, a **PDA** has **7**, and a **TM** has **7**.
- A **Moore/Mealy** machine has **6** components.
- A **CFG** has **4** components, a **regular expression** alphabet has **3** operators.
- **CNF** uses $2n-1$ steps; **GNF** uses n steps for a length- n string.
- **CYK** is $O(n^3)$; **Hopcroft** is $O(n \log n)$; subset construction is $O(2^n)$.
- The **Chomsky hierarchy** has **4** types, mapping to **4** machines and **4** language classes.

True/False recall (deep)

- "**NFA** is more powerful than **DFA**" is **FALSE** (they are equal).
- "**NPDA** is more powerful than **DPDA**" is **TRUE**.
- "**Regular** languages are closed under **intersection**" is **TRUE**.
- "**CFLs** are closed under **intersection**" is **FALSE**.
- "**CFLs** are closed under **complement**" is **FALSE**.
- "**RE** languages are closed under **complement**" is **FALSE**.
- "**Recursive** languages are closed under **complement**" is **TRUE**.
- "Every **context-free** language is **recursive**" is **TRUE**.
- "The **Halting problem** is **decidable**" is **FALSE**.
- " $a^n b^n c^n$ is **context-free**" is **FALSE** (it is context-sensitive).
- "The **pumping lemma** proves a language **IS** regular" is **FALSE** (it only proves non-regularity).
- "**Myhill-Nerode** is both necessary and sufficient for regularity" is **TRUE**.
- "CFG **equivalence** is **decidable**" is **FALSE**.
- "DFA **equivalence** is **decidable**" is **TRUE**.
- "**L*** always contains **epsilon**" is **TRUE**.
- " $\phi^* = \phi$ " is **FALSE** (it equals $\{\epsilon\}$).

Mixed atomic facts (deep)

- A **transition diagram** marks the start state with **1** incoming arrow and final states with **double circles**.
- The **yield** of a parse tree is read **left to right** across the **leaves**.
- **Arden's theorem** is used in the **FA-to-RE** conversion.
- **Thompson's construction** is used in the **RE-to-NFA** conversion.
- A **universal language** L_u is **RE** but **not recursive**.
- **Diagonalization** is the technique proving the **Halting problem** undecidable.
- The **stack** of a PDA follows **LIFO** (last-in, first-out) discipline.
- A **2-stack PDA** is as powerful as a **Turing Machine**.
- A **queue automaton** (PDA with a queue) is also equivalent to a **Turing Machine**.
- A **counter machine** with **2** counters is equivalent to a **Turing Machine**.
- A **1-counter** automaton is strictly **weaker** than a **PDA**... it is a special case recognising some non-regular languages.
- The minimal DFA's state count equals the **Myhill-Nerode** index.
- Two DFAs are **equivalent** iff their **minimal** forms are **isomorphic**.
- A **regular** language always has **both** a left-linear and a right-linear grammar.
- The **reverse** of a regular language is **regular**.
- The **reverse** of a CFL is a **CFL**.
- A homomorphism replaces each **symbol** by a **string**, preserving **regularity** and **context-freeness**.
- **Closure** under **union** holds for **all** four major classes (Regular, CFL, Recursive, RE).
- **Closure** under **intersection** fails only for **CFL** among the four major classes.
- The **empty stack** acceptance mode ignores **final states** entirely.
- An **epsilon-NFA** with **n** states still converts to a DFA with at most 2^n states.
- **Kleene's theorem** and **Myhill-Nerode** both characterise **regular** languages.

Examiner-favourite numeric facts

- A binary-divisible-by-**3** DFA needs exactly **3** states.
- A "**contains 101**" DFA needs exactly **4** states (minimal).

- A "**third symbol from start is a**" DFA needs a constant number of states, around **5** including dead.
- The number of **distinct** DFAs (up to renaming) for a language is **1** minimal one.
- A **2-state** DFA can recognise "**ends in a**" over **{a,b}**.
- The **Cook-Levin** theorem dates to **1971**; Turing's halting proof to **1936**.
- **SAT** was the **1st** problem proven **NP-complete**.
- There are **3** resource classes commonly compared: **P**, **NP**, and **NP-complete**.

Final consolidation

- The **4** questions most asked in CIL MT TOC: **class identification**, **closure true/false**, **minimal-DFA counting**, and **decidability**.
- **Regular** = **finite memory**; **CFL** = **stack memory**; **CSL** = **linear memory**; **RE** = **unbounded memory**.
- The **single most-tested** contrast: **equivalence** is decidable for **regular** but not for **CFL**.
- The **second most-tested** contrast: **NFA=DFA** but **NPDA>DPDA** in power.
- The **third most-tested** contrast: **Recursive** closed under complement, **RE** not.
- **aⁿ bⁿ** (CFL), **aⁿ bⁿ cⁿ** (CSL), **ww** (non-CFL): the **3** canonical separating languages.
- A **TM** is the **1** model defining **algorithmic** computability via the **Church-Turing** thesis.
- Memorise the **4x** master table: **class -> machine -> grammar -> closure -> decidability** for full marks.

14. Additional Atomic One-Liners (Set 2)

Foundations extended

- An **alphabet** must exclude **0** infinite-size possibilities since it is by definition **finite**.
- The number of strings of length exactly **2** over $\{a,b\}$ is **4**: **aa, ab, ba, bb**.
- The number of strings of length exactly **3** over $\{0,1\}$ is **8** (2^3).
- The concatenation operation is **closed** on Σ^* , giving exactly **1** resulting string per pair.
- The **prefix closure** of a language adds all **prefixes** of its members.
- A **proper substring** of **w** excludes **2** cases: **w** itself and (commonly) **epsilon**.
- $(w^R)^R$ equals exactly **1** string: **w** (double reversal is identity).
- The **power** $a^0 b^0$ equals **epsilon**, of length **0**.
- A language over a **1**-symbol (unary) alphabet is called a **unary language**.
- Every **unary** context-free language is in fact **regular**, a **1**-of-a-kind collapse.

DFA extended

- A DFA accepting **exactly** the string **ab** needs **4** states including a **dead** state.
- The number of **final** states in a DFA can be **0**, meaning it accepts **phi**.
- A DFA with **all** states final still rejects **0** strings only if it is **complete** with no dead state.
- A DFA's language is **infinite** iff its transition graph has a **reachable** cycle leading to a **final** state.
- The **product** DFA for **difference L1 - L2** accepts when DFA1 is **final** and DFA2 is **non-final**.
- A DFA recognising "**starts with a**" over $\{a,b\}$ needs **3** states (incl. dead).
- A DFA for "**contains an even number of 0s and odd number of 1s**" needs **4** states (2×2 product).
- The **empty** DFA (no states) is typically **disallowed**; a DFA needs at least **1** state.
- The **trap** state contributes **0** accepted strings.
- Counting the minimal DFA for "**divisible by 6**" gives **6** states.

NFA / conversion extended

- An NFA may have **multiple** start states in some textbook variants, normalised to **1** via **epsilon**.
- The subset construction's reachable-states count is $\leq 2^n$ but often far **fewer**.
- An NFA accepting "**ends with ab**" needs **3** states; its DFA needs **3** states too (no blow-up here).
- The **worst-case** blow-up example is "**n-th symbol from end**", needing 2^n DFA states.
- Converting an **epsilon-NFA** first computes **epsilon-closure** of **1** start state.
- A DFA state in subset construction is **dead** if it equals the **empty set**.

Minimization extended

- Two final states are always **indistinguishable** by the **empty** string, needing a longer **witness**.
- The minimization **partition refinement** starts with **2** groups: **final** and **non-final**.
- Each refinement step can only **increase** or keep the number of **partitions**.
- Minimization terminates when **no** partition splits further, giving the **minimal** DFA.
- The minimal DFA is **unique** up to **state renaming (isomorphism)**.

Regular expression extended

- The RE **a+** equals **aa***, denoting **1** or more **as**.

- The RE **a?** (optional) equals **epsilon + a**, denoting **0 or 1 a**.
- The RE **(ab)*** denotes strings that are **repetitions** of the block **ab**.
- The RE **a*(b a*)*** can describe strings over **{a,b}** with **any** arrangement.
- **Two** REs denoting the same language are **equivalent**, decided via **minimal DFA** comparison.
- The RE for "**even number of a's**" over **{a}** is **(aa)***.
- The RE for "**strings not containing 11**" uses **(0+10)*(1+epsilon)**.

Closure extended

- **Regular** is closed under **init/prefix**, **suffix**, and **cycle** operations.
- **Regular** is closed under **max** and **min** operations on languages.
- **CFL** is closed under **cyclic shift**.
- **CFL** $\cap$ **CFL** may be **non-CFL**, shown by **aⁿ bⁿ cⁿ**.
- **Recursive** $\cap$ **Recursive** is always **recursive**.
- **RE** $\cap$ **RE** is always **RE**; **RE complement** may **not** be RE.

Pumping extended

- For a regular language, the **pump** repeats **1** block **y** any number of times.
- For a CFL, the **pump** repeats **2** blocks **v** and **x** in lockstep.
- The pumping lemma is a **game**: the adversary picks **p**, you pick **w**, adversary splits, you pick **i**.
- To break regularity you choose **1** value of **i** (often **i=2** or **i=0**) putting the string **outside L**.
- **aⁿ bⁿ** fails regular pumping because pumping **as** breaks the **equality** of counts.
- **aⁿ bⁿ cⁿ** fails CFL pumping because **vw^x** cannot span all **three** blocks.

CFG / PDA extended

- **Leftmost** and **rightmost** derivations of an **unambiguous** grammar give the **same** parse tree.
- A **derivation step** $\Rightarrow^*$ denotes **zero or more** rewriting steps.
- **Useless symbol** removal can shrink a grammar to **0** productions if **L** is empty.
- A **nullable** variable derives **epsilon** in **≥ 1** step.
- **CNF** forbids **epsilon**-productions except possibly **S $\rightarrow$ epsilon**.
- A **DPDA** has **at most 1** applicable move per configuration.
- The **2-stack** PDA jumps to **full** Turing power.
- **NPDA** accepts **{ww^R}**; **DPDA** cannot, a **1-language** power gap demonstration.
- A **PDA** can be built from a **CNF** grammar with a **single** state.
- The **height** of a PDA stack can be **unbounded**, unlike the **0** memory of an FA.

CFL decision extended

- **CYK** fills a triangular table of **n(n+1)/2** cells in **O(n³)** time.
- CFL **membership** is decidable; CFL **equivalence** is **undecidable**.
- CFL **emptiness** reduces to checking if **S** is a **generating** symbol.
- CFL **finiteness** reduces to **cycle detection** in a cleaned grammar's graph.
- "Is a CFL equal to **Sigma***?" is **undecidable**.

TM extended

- A TM tape extends to **infinity** in at least **1** direction.

- A **multi-tape** TM with **k** tapes simulates in $O(t^2)$ on **1** tape.
- A **non-deterministic** TM with time **t** simulates deterministically in $2^{O(t)}$ time.
- The **UTM** demonstrates a **single** machine can run **any** program.
- An **enumerator** for a **recursive** language outputs strings in **canonical** order.
- A TM that **halts** on all inputs decides a **recursive** language.
- The **busy beaver** function is **non-computable**, growing faster than any **computable** function.
- A **linear bounded automaton** uses $O(n)$ tape cells.

Computability extended

- **Recursive** $\subset$ **RE** with the gap witnessed by **A_TM**.
- A language and its complement both **RE** implies the language is **recursive** (Post).
- **Co-RE** is the class of **complements** of RE languages.
- **A_TM complement** is in **co-RE** but **not** in **RE**.
- The class **RE** $\cap$ **co-RE** equals exactly the **recursive** languages.

Decidability extended

- "Does a TM accept ≥ 1 string?" is **undecidable** (non-emptiness, by Rice).
- "Does a TM accept a **regular** language?" is **undecidable** (Rice).
- "Does a TM accept a **context-free** language?" is **undecidable** (Rice).
- "Does a TM run for **>100** steps on input w?" is **decidable** (just simulate).
- "Is the language of a DFA **infinite**?" is **decidable**.
- "Do two **TMs** accept the same language?" is **undecidable**.
- "Is a TM's language **empty**?" is **undecidable** (**E_TM**).
- "Is **w** accepted by TM **M** within **k** steps?" is **decidable**.
- **Rice's theorem** covers **all** non-trivial **semantic** properties, totalling **0** decidable ones among them.

Complexity extended

- **P** is contained in **NP**, which is contained in **PSPACE**.
- **NP** problems have **polynomial-size** certificates verifiable in **polynomial** time.
- A problem is **NP-complete** if it is in **NP** and **NP-hard**.
- **TSP** (decision version) is **NP-complete**; the **optimization** version is **NP-hard**.
- **P = NP** is **unresolved** and worth a **1-million-dollar** prize.
- **Cook-Levin** reduced every NP problem to **SAT**.
- **Reductions** used in NP-completeness run in **polynomial** time.

Master recall (Set 2)

- The **4** closure facts to memorise: Regular **all**, CFL **not** $\cap$ **complement**, Recursive **all incl complement**, RE **not complement**.
- The **3** power facts: **NFA=DFA**, **NPDA>DPDA**, **NTM=DTM**.
- The **4** machine-tuple sizes: **DFA=5**, **Moore/Mealy=6**, **PDA=7**, **TM=7**.
- The **2** step-count formulas: **CNF=2n-1**, **GNF=n**.
- The **3** algorithm complexities: **CYK=O(n^3)**, **Hopcroft=O(n log n)**, **subset=O(2^n)**.
- The **3** undecidable CFL problems: **equivalence**, **ambiguity**, **totality (L=Sigma*)**.
- The **2** historical dates: **Halting 1936**, **Cook-Levin 1971**.
- The **4** canonical non-regular/non-CFL languages: **a^n b^n**, **a^n b^n c^n**, **ww**, **a^p (prime)**.

- The **1** theorem that is necessary AND sufficient for regularity: **Myhill-Nerode**.
- The **1** model defining computability: the **Turing Machine** via **Church-Turing**.
- The **4**-row master table: each class maps to a **machine**, **grammar**, **closure profile**, and **decidability profile**.
- **Regular** $\subset$ **DCFL** $\subset$ **CFL** $\subset$ **CSL** $\subset$ **Recursive** $\subset$ **RE** captures **6** nested classes.
- **DCFL** sits strictly between **Regular** and **CFL**, recognised by a **DPDA**.
- The **empty stack** and **final state** PDA modes coincide for **NPDA** but **not** for **DPDA**.
- **Greibach Normal Form** removes **left recursion** as a side effect.
- A grammar with production **S $\rightarrow$ aSb** alone generates **$\{a^n b^n : n \geq 1\}$** ... adding **S $\rightarrow$ ab** or **epsilon** sets the base.
- The **language** accepted by a minimal DFA and its source DFA are **identical**.

15. Final Atomic One-Liners (Set 3)

- A **symbol** is the **1** smallest indivisible unit of an alphabet.
- The **Kleene closure** of $\{a\}$ gives $\{\epsilon, a, aa, aaa, \dots\}$, an **infinite** set.
- A language with exactly **0** strings is **ϕ** ; with exactly **1** string **ϵ** is $\{\epsilon\}$.
- A **DFA** reading **ϵ** stays in **1** place: the **start state**.
- An NFA's **δ^*** returns a **set** of states, possibly the **empty** set.
- The **ϵ -closure** of a set is the **union** of closures of its **members**.
- A **minimal** DFA has **0** redundant (equivalent) states.
- **Moore** machines output even before reading **1** symbol (the start-state output).
- A **regular expression** and a **DFA** describe the **same** regular language by **Kleene's theorem**.
- The **Arden** solution **QP^*** is unique iff **P** lacks **ϵ** .
- **Regular** languages are closed under **symmetric difference** (a Boolean operation).
- The **pumping** constant equals the DFA's **state** count.
- **Myhill-Nerode** classes partition **Σ^*** into **finitely** many groups iff **L** is regular.
- A **CFG** can be ambiguous while its **language** has an **unambiguous** grammar.
- **Inherent ambiguity** means **every** grammar for the language is **ambiguous**.
- **CNF** restricts RHS to **2** variables or **1** terminal.
- **GNF** restricts RHS to **1** leading terminal followed by **variables**.
- A **PDA** stack uses **LIFO** order with initial symbol **Z_0** .
- **NPDA** = **CFL** and **DPDA** = **DCFL**, with **DCFL** $\subset$ **CFL**.
- **CFL** $\cap$ **Regular** stays **CFL**, the **1** intersection that is safe.
- **CYK** requires the grammar in **CNF** and runs in **$O(n^3)$** .
- A **TM** accepts, rejects, or **loops**, totalling **3** possible fates.
- **Recursive** = **decidable** = TM that **always halts**.
- **RE** = **semi-decidable** = TM that **halts on acceptance**.
- **Recursive** $\subset$ **RE**, with **A_TM** in **RE** but not **recursive**.
- The **Halting problem** reduces to most **undecidable** TM problems.
- **Rice's theorem** makes all **non-trivial language** properties **undecidable**.
- **PCP** is **undecidable** but its **unary** restriction is **decidable**.
- The **universal language** **L_u** is **RE**, its **complement** is **not RE**.
- **P** is **deterministic polynomial**; **NP** is **non-deterministic polynomial**.
- **SAT** is **NP-complete** by **Cook-Levin (1971)**.
- **NP-hard** need not be **in NP**, e.g. **Halting problem**.
- The hierarchy of memory: **none** $<$ **stack** $<$ **linear-tape** $<$ **unbounded-tape**, matching **FA, PDA, LBA, TM**.
- A **2-counter machine** matches **Turing** power, surprising for just **2** counters.
- **Context-sensitive** languages are closed under **complement** (Immerman-Szelepcsenyi, 1987).
- **Regular** languages have a **unique** minimal DFA but **many** equivalent NFAs.
- The **reversal** of a regular language is recognised by **reversing** all DFA edges and swapping **start/final**.
- **Thompson's construction** adds **ϵ -edges** for **union, concatenation, star**.
- A **dead state** is needed to **complete** a partial DFA before **complementation**.
- The **3** most common CIL MT TOC question types: **construct, classify, closure/decidability true-false**.
- **$a^n b^n$** needs **1** stack; **$a^n b^n c^n$** needs **2** comparisons hence an **LBA**.
- **Σ^*** is **countable**; the set of all **languages** is **uncountable**, forcing some **non-RE** languages.
- The **minimal** number of states to recognise **ϕ** is **1** (a single dead state).

- The **minimal** number of states to recognise **Sigma*** is **1** (a single final state).
- The **Chomsky** mapping to memorise: **Type 3=FA, Type 2=PDA, Type 1=LBA, Type 0=TM.**